CSAF-Net: A Cross-Attention Gated Fusion Network for CNN–Vision Transformer Based Concrete Crack Detection

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***Abstract*—Concrete crack detection is an important task for structural inspection and maintenance. This paper presents CSAF-Net, a hybrid deep learning architecture that combines DenseNet-121 and Vision Transformer (ViT) features through bidirectional cross-attention and learned gated fusion. The model is evaluated on a balanced 4,000-image subset of SDNET2018. Experiments compare CNN-only, ViT-only, feature concatenation, ungated cross-attention, scalar-gated fusion, and the proposed gated fusion. The full model achieves 86.00% test accuracy, 87.00% recall, and 86.14% F1-score in the original run. Repeated runs are also examined to assess variability. A separate synthetic wide-view localization evaluation highlights the challenges of locating cracks in larger images.**

***Index Terms*—Concrete crack detection, computer vision, DenseNet-121, Vision Transformer, cross-attention, gated fusion.**

I. Introduction

Cracks in concrete structures can indicate deterioration and may compromise structural durability if they remain undetected. Automated image-based crack detection offers a practical approach to inspecting concrete surfaces. However, crack appearance varies with lighting, surface texture, crack width, and background patterns, making reliable classification challenging.

Convolutional neural networks (CNNs) learn local visual features that are useful for identifying crack patterns. DenseNet-121 uses dense connections to facilitate feature reuse and information propagation [1]. Vision Transformers (ViTs), in contrast, model relationships between image patches through self-attention [2], [3]. These complementary properties motivate the investigation of hybrid CNN–Transformer architectures for concrete crack detection.

This paper presents CSAF-Net, a CNN–Vision Transformer architecture that combines DenseNet-121 and ViT-Base/16 using bidirectional cross-attention and learned gated fusion. The proposed design projects both feature streams into a shared representation space, enables information exchange between the streams, and learns their relative contributions for binary crack classification.

The main contributions of this work are: 1. A dual-backbone architecture combining CNN-based local features and Transformer-based contextual representations. 2. A bidirectional cross-attention module followed by token-wise, channel-wise gated feature fusion. 3. An ablation study comparing individual backbones and alternative fusion strategies. 4. An exploratory wide-view localization evaluation that examines the limitations of applying a patch classifier to larger concrete images.

II. Literature Survey

Prior work on concrete crack detection includes image-based classification and segmentation approaches. SDNET2018 provides annotated concrete surface images for crack detection, while CrackVision investigates transfer learning for concrete crack classification [4], [5]. Attention-based methods have also been applied to concrete crack analysis and reinforced-concrete bridge inspection [6], [7].

CNNs such as DenseNet extract local texture and edge information, whereas Vision Transformers model relationships among image patches [1]–[3]. Hybrid CNN–Transformer models, cross-attention, and gated feature-fusion methods have also been investigated in other image-analysis domains [8]–[11]. Related research covers solar-cell and tool-wear defect classification, lightweight attention, medical-image preprocessing, eye-disease diagnosis, and industrial defect detection [12]–[19].

Table I summarizes selected studies, their application domains, methods, and relevance to CSAF-Net. Because these studies address different datasets and tasks, their findings provide context rather than direct performance comparisons with the proposed model.

III. Methods and Materials

A. Overall Architecture

CSAF-Net combines convolutional feature extraction and transformer-based context modeling through bidirectional cross-attention and learned gated fusion. The architecture consists of two pretrained backbones, feature projection layers, a cross-attention fusion module, and a binary classification head. DenseNet-121 extracts local visual representations, while ViT-Base/16 captures relationships among image patches.

Given an input image $I \in \mathbb{R}^{3\times224\times224}$, both backbones produce feature tokens that are projected into a shared 256-dimensional space. The pretrained backbone parameters are

TABLE I
LITERATURE SURVEY AND RELEVANCE TO CSAF-NET

Study	Domain	Method	Finding and Relevance
Abdelsattar et al. (2025)	Solar-cell defect detection	Deep CNN classifiers	Demonstrates automated visual defect classification; results can be sensitive to preprocessing and class imbalance.
Alkannad et al. (2025)	Concrete crack detection	CrackVision; pretrained CNN transfer learning	Provides a direct concrete-crack classification baseline using transfer learning.
Alnamoly et al. (2024)	Tomato-leaf disease classification	FL-ToLeD; lightweight attention-augmented CNN	Shows the use of attention to improve fine-grained visual classification.
Eisentraut et al. (2025)	Tuberculosis detection from chest X-rays	CNN with Gaussian-filter preprocessing	Highlights the role of preprocessing in image-classification performance.
Feng et al. (2025)	Tool-wear classification	Improved DenseNet	Provides a related application of densely connected features for fine-grained classification.
Hajian and Aramvith (2025)	Remote-sensing super-resolution	AERU-Net with edge recovery and attention	Provides a related example of attention-based feature refinement and learned gating.
Hussain and Shouno (2024)	Medical-image segmentation	MAGRes-UNet; multi-attention gated residual U-Net	Provides a precedent for gated feature integration in visual recognition.
Lee et al. (2025)	MRI-guided surgical planning	Multi-view cross-attention	Demonstrates cross-attention for exchanging information across feature views.
Lu et al. (2022)	Internal defect detection in magnetic tiles	MMFCNN with cross-attention fusion	Provides a related industrial-defect example of cross-attention-based feature fusion.
Rashid et al. (2025)	Tomato-leaf disease detection	MobileNetV3 with inverted residuals and channel attention	Illustrates channel attention in lightweight visual classification.
Rieck et al. (2025)	Eye-disease diagnosis	Transformer–CNN hybrid	Provides cross-domain evidence for hybrid convolutional and Transformer representations.
Ruggieri et al. (2025)	Reinforced-concrete bridge inspection	Attention-augmented defect detector	Directly relates attention mechanisms to structural-defect inspection.
Sarhadi et al. (2024)	Concrete-crack segmentation	Attention-based Swin U-Net	Studies attention-based crack segmentation, a related task with a different output objective.
Venugopal et al. (2026)	Thyroid diagnosis from ultrasound	Multi-scale Vision Transformer with neural architecture search	Provides an indirect example of Transformer architecture optimization in medical imaging.
Zhang et al. (2025)	Tomato-disease recognition	Image dehazing with improved MobileNetV3	Provides a related example of preprocessing before image classification.

frozen during training, while the projection layers and subsequent modules are optimized for crack classification.

B. Feature Extraction and Projection

The DenseNet-121 feature extractor produces a feature map of size $1024 \times 7 \times 7$. Flattening its spatial dimensions produces 49 tokens, each with 1024 channels. A linear projection maps these tokens into a 256-dimensional feature space:

$$X_c = \mathrm{LN}(W_c T_c + b_c), \tag{1}$$

where $T_c \in \mathbb{R}^{49\times1024}$ denotes the flattened CNN features, W_c and b_c are learnable projection parameters, and LN denotes layer normalization.

The ViT-Base/16 backbone produces one classification token and 196 patch tokens, each with 768 dimensions. The classification token is removed, and the remaining patch tokens are projected as

$$X_v = \mathrm{LN}(W_v T_v + b_v), \tag{2}$$

where $T_v \in \mathbb{R}^{196\times768}$ contains the patch embeddings. Both projected representations have 256 feature dimensions.

C. Bidirectional Cross-Attention

Cross-attention enables each feature branch to incorporate information from the other branch. Let X_c and X_v denote the projected CNN and ViT token sequences, respectively. The CNN-to-ViT attention operation uses CNN tokens as queries and ViT tokens as keys and values:

$$A_c = \mathrm{LN}\left(\mathrm{MHA}(X_c, X_v, X_v) + X_c\right). \tag{3}$$

In the reverse direction, ViT tokens query the CNN representations:

$$A_v = \mathrm{LN}\left(\mathrm{MHA}(X_v, X_c, X_c) + X_v\right). \tag{4}$$

Both attention modules use eight heads, residual connections, and layer normalization. The resulting sequences contain 49 and 196 tokens, respectively. To align them for fusion, the

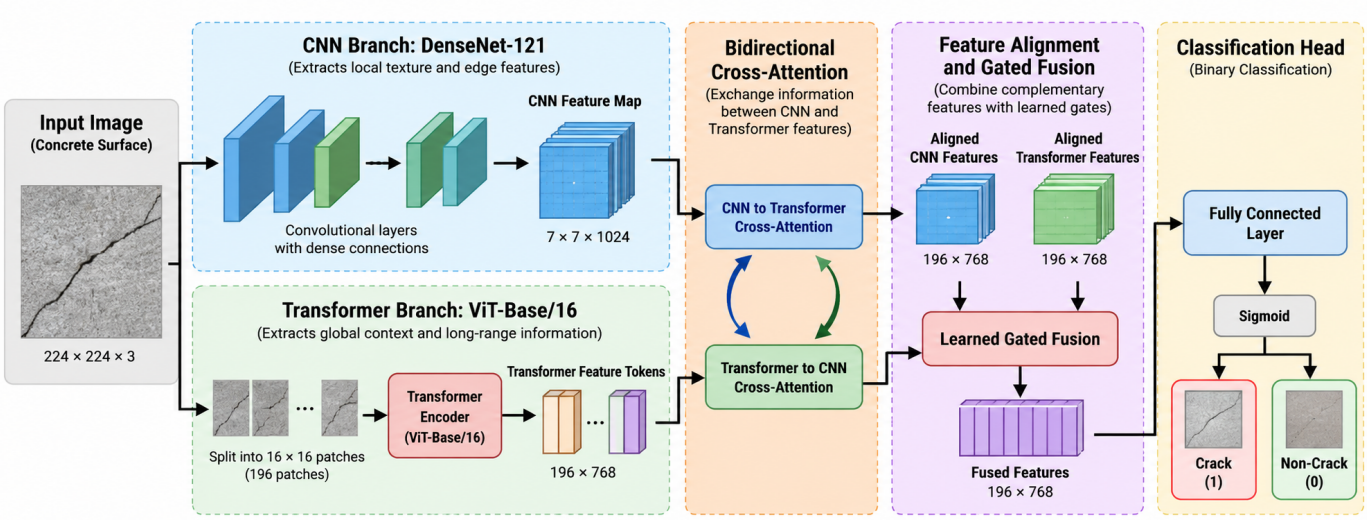


Fig. 1. Overview of CSAF-Net. DenseNet-121 and ViT-Base/16 extract complementary features that interact through bidirectional cross-attention. The aligned representations are combined by a learned gate before binary crack classification.

ViT-attended sequence is adaptively average-pooled along the token dimension to produce 49 tokens. Denote the aligned branches by $\widetilde{A}_c$ and $\widetilde{A}_v$.

D. Learned Gated Fusion

The aligned representations are concatenated along the feature dimension and passed through a two-layer gating network:

$$G = \sigma\left(W_2\,\mathrm{ReLU}\left(W_1[\widetilde{A}_c;\widetilde{A}_v]+b_1\right)+b_2\right), \tag{5}$$

where the gating network has dimensions $512 \to 256 \to 256$. The sigmoid function constrains gate values to the interval $[0,1]$. The gate is input-conditioned and varies across tokens and feature channels.

The final fused representation is

$$F = G \odot \widetilde{A}_c + (1-G)\odot \widetilde{A}_v, \tag{6}$$

where $\odot$ denotes element-wise multiplication. This formulation allows the network to assign different weights to the two attended representations at each token and channel.

E. Classification Head

The fused token sequence is mean-pooled across its 49 tokens to obtain a 256-dimensional vector:

$$z = \frac{1}{49}\sum_{i=1}^{49} F_i. \tag{7}$$

The pooled vector is processed by a fully connected layer with 128 hidden units, ReLU activation, dropout with probability 0.3, and a final linear layer that produces a single logit. Binary cross-entropy with logits is used as the training objective. The sigmoid function converts the output logit to a probability during inference.

IV. Experimental Setup

A. Dataset and Data Partitioning

Experiments were conducted using the wall subset of SDNET2018, which contains images of cracked and uncracked concrete surfaces. A balanced subset of 4,000 images was used, comprising 2,000 images per class. The dataset was partitioned into training, validation, and test sets using a 70:15:15 ratio. The training set contained 2,800 images, while the validation and test sets contained 600 images each. Each partition was balanced, with equal numbers of cracked and uncracked images.

Representative examples from the dataset used in this study are shown in Fig. 3, illustrating concrete surface images from the crack and non-crack classes.



Fig. 3. Representative dataset images. Top: non-crack examples; bottom: crack examples.

The dataset preparation script used Python's random-number generator with seed 42 and a maximum selection limit of 2,000 images per class. The existing dataset split was retained for the reported experiments.

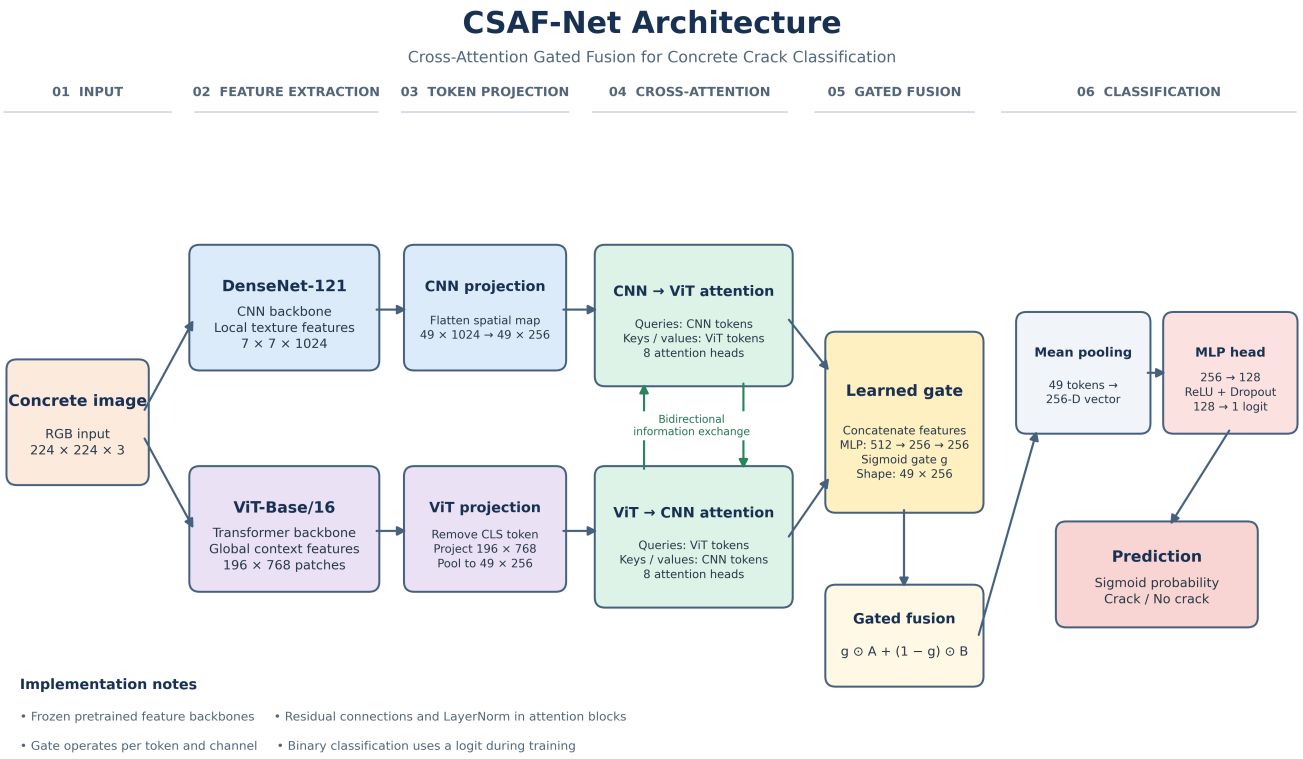


Fig. 2. Technical architecture of CSAF-Net, showing the DenseNet-121 and ViT-Base/16 branches, bidirectional cross-attention, learned gated fusion, and binary classification head.

B. Image Preprocessing

All input images were resized to 224×224 pixels. Images were converted to tensors and normalized using the ImageNet channel-wise mean $(0.485, 0.456, 0.406)$ and standard deviation $(0.229, 0.224, 0.225)$. Random horizontal flipping was applied to training images only. Validation and test images underwent resizing and normalization without random augmentation.

C. Training Configuration

The models were trained for five epochs with a batch size of eight. AdamW was used as the optimizer with a learning rate of 10^{-4}. Binary cross-entropy with logits served as the loss function. The DenseNet-121 and ViT backbone parameters were frozen, while the projection layers, fusion module, and classification head were trainable. The checkpoint with the highest validation F1-score was saved and reloaded for evaluation on the held-out test set.

The experiments were implemented in Python 3.9.6 using PyTorch 2.8.0 and Torchvision 0.23.0. The development environment reported availability of Apple's Metal Performance Shaders (MPS) backend.

D. Evaluation Metrics

Classification performance was evaluated using accuracy, precision, recall, and F1-score. Accuracy measures the proportion of correctly classified images. Precision measures the proportion of predicted crack-positive images that are actually cracked, while recall measures the proportion of cracked images correctly detected. F1-score is the harmonic mean of precision and recall.

These metrics are reported for all six classification variants to compare the individual backbones and the contribution of the fusion strategies.

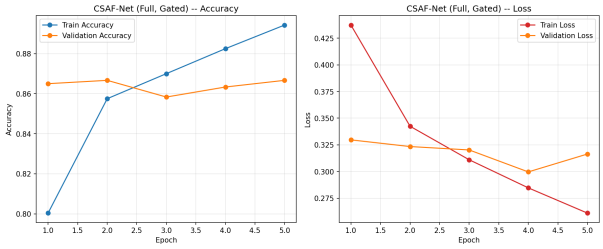


Fig. 4. Training and validation curves for the proposed CSAF-Net model.

V. Results and Discussion

A. Classification Ablation Study

Table II presents the test-set performance of six model variants. The CNN-only and ViT-only models provide single-backbone baselines. Feature concatenation, ungated cross-attention, and scalar-gated fusion are evaluated to investigate alternative feature-combination strategies.

TABLE II
CLASSIFICATION RESULTS ON THE HELD-OUT TEST SET

Model	Acc.	Prec.	Rec.	F1
CNN-only	75.33	79.46	68.33	73.48
ViT-only	82.83	85.82	78.67	82.09
Concatenation	84.17	84.51	83.67	84.09
Ungated cross-attention	85.83	89.09	81.67	85.22
Scalar-gated fusion	85.67	86.64	84.33	85.47
Full CSAF-Net	**86.00**	85.29	**87.00**	**86.14**

All metrics are percentages.

The full CSAF-Net model achieved 86.00% accuracy, 87.00% recall, and an F1-score of 86.14% in the original run. Compared with the CNN-only baseline, the F1-score increased by 12.66 percentage points. The ViT-only baseline also outperformed the CNN-only model, indicating that transformer features provide useful information for this classification task.

Feature concatenation achieved an F1-score of 84.09%, while ungated cross-attention and scalar-gated fusion achieved 85.22% and 85.47%, respectively. The full model produced the highest F1-score among the six original runs, although ungated cross-attention achieved the highest precision at 89.09%.

These results show that the proposed combination of bidirectional cross-attention and learned gating is promising for crack classification. However, the single-run ablation results alone do not establish that the learned gate produces a statistically significant improvement over alternative fusion strategies.

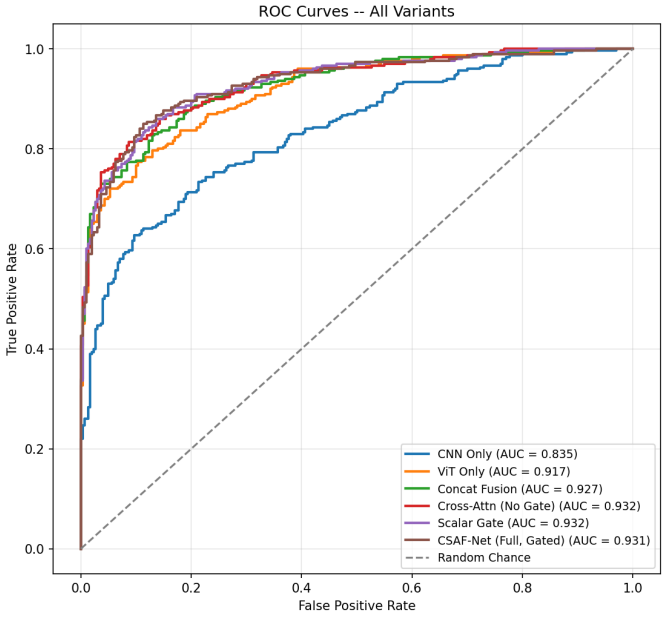


Fig. 5. Receiver operating characteristic (ROC) curves for the evaluated classification variants.

B. Repeated-Run Analysis

To examine variability, the full CSAF-Net, scalar-gated fusion, and ungated cross-attention variants were evaluated in three runs each, consisting of the original run and two additional reruns. Table III reports the mean and sample standard deviation.

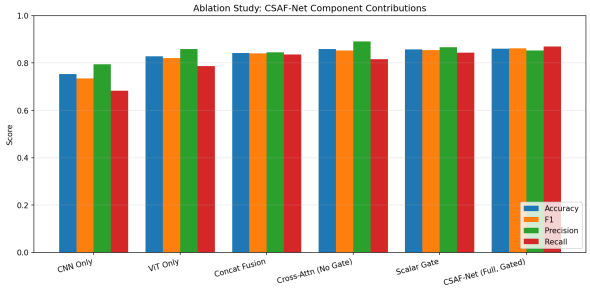


Fig. 6. Comparison of accuracy, F1-score, precision, and recall across the six classification variants.

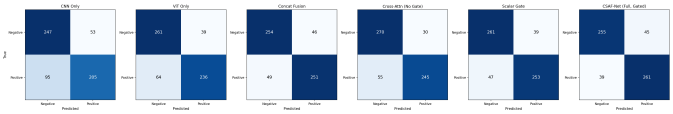


Fig. 7. Confusion matrices for the evaluated classification variants on the held-out test set.

TABLE III
REPEATED-RUN CLASSIFICATION RESULTS

Model	Acc.	Prec.	Rec.	F1
Full CSAF-Net	85.78 ± 0.54	85.64 ± 1.26	86.00 ± 1.00	85.81 ± 0.46
Scalar-gated	85.94 ± 0.63	87.09 ± 2.87	84.56 ± 2.34	85.75 ± 0.30
Ungated cross-attn.	86.06 ± 0.38	89.24 ± 0.37	82.00 ± 1.20	85.46 ± 0.52

Values are mean $\pm$ sample standard deviation in percentages over three runs.

Across the three runs, full CSAF-Net had the highest mean recall and a marginally higher mean F1-score than the other two fusion variants. Ungated cross-attention achieved the highest mean precision and accuracy. The differences in mean F1-score were small, so these results should be interpreted as observed trends rather than evidence of statistically significant superiority. The original run's random seed was not recorded, and only three runs were available for each variant.

C. Wide-View Crack Localization

In addition to image-level classification, a separate synthetic wide-view benchmark was used to investigate whether the classification model could localize cracks within larger images. The benchmark used 900×900 scenes containing either a crack region or no crack region. Crack regions were generated with sizes between 150 and 300 pixels. A sliding-window procedure used 224×224 crops with a stride of 56 pixels. A predicted window was considered localized when its intersection over union (IoU) with the reference box exceeded 0.3.

The detection threshold was selected using validation scenes under a constraint of at most 0.2 false-positive boxes per clean scene. The selected threshold for the single CSAF-Net model was 0.995. On the held-out synthetic test benchmark, the model localized 45 of 100 crack-containing scenes, giving a scene-level localization recall of 45% (95% Wilson confidence

interval: approximately 35.6–54.8%). The mean IoU among localized cracks was 0.541, with 0.14 false-positive boxes per clean scene. Scene-level precision was 80.4% and scene-level F1-score was 57.7%.

A secondary threshold analysis evaluated a threshold of 0.992 on the same held-out scenes. It localized 52 of 100 crack scenes (52%), with mean IoU of 0.536 and 0.52 false-positive boxes per clean scene. Although recall increased, the false- positive rate exceeded the predefined validation budget, and scene-level F1-score decreased to 55.3%. Therefore, the validation-selected threshold of 0.995 is retained as the primary result; the 0.992 analysis is reported only as a secondary operating-point comparison.

These findings indicate that high image-level classification performance does not automatically translate into reliable localization in wide-view images. The synthetic benchmark is limited in its scene construction and should not be interpreted as evidence of equivalent performance on unrestricted real-world images.

VI. Limitations

This study has several limitations. First, the classification experiments use a balanced subset of SDNET2018 wall images. Evaluation on additional datasets and real-world acquisition conditions is necessary to establish cross-domain generalization.

Second, the repeated-run analysis includes only three runs for each of the three fusion variants. The observed differences in mean F1-score are small and do not establish statistically significant superiority of the proposed gating mechanism. Additional repeated experiments and controlled statistical comparisons would provide stronger evidence.

Third, the wide-view localization benchmark uses synthetic 900×900 scenes and predefined crack regions. Its results may not reflect the complexity of real structures, including variable lighting, surface texture, perspective, and crack geometry. The localization pipeline relies on sliding-window classification rather than a model trained explicitly for object detection or pixel-level segmentation. Consequently, the reported localization performance should be interpreted within the scope of the synthetic benchmark.

Finally, the dataset preparation script fixes the random seed but does not sort the source filenames before shuffling. Consequently, reproducing the exact split may depend on the ordering returned by the file system. The script does not save a split manifest; therefore, the generated dataset and its file assignments should be retained to reproduce the reported experiments.

VII. Conclusion

This paper presented CSAF-Net, a hybrid architecture that combines DenseNet-121 and ViT-Base/16 representations using bidirectional cross-attention and learned gated feature fusion for concrete crack classification. On the evaluated 4,000-image SDNET2018 wall subset, the original run achieved 86.00% test accuracy, 87.00% recall, and an F1-score of 86.14%. Among the six original classification variants, the full model achieved the highest accuracy, recall, and F1-score in that run.

Across repeated runs, the full model achieved the highest mean recall and a marginally higher mean F1-score than the scalar-gated and ungated cross-attention variants. These differences do not establish statistically significant superiority over the alternatives.

The separate synthetic wide-view evaluation revealed a remaining challenge: image-level classification performance does not guarantee reliable crack localization in larger scenes. Future work will focus on broader cross-dataset evaluation, additional repeated experiments, improved localization methods, and validation on real-world images with reliable ground-truth annotations.

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